Sayyad Yusuff Basha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.61
LAKSHYA SCHOOL	Class XII (CBSE)	2023	91.80%
ADITYA (EM) HIGH SCHOOL	Class X (ANDHRA STATE BOARD)	2021	100.00%

SKILLS

Languages: C++, Python, JavaScript, SQL

Machine Learning: Scikit-learn, Pandas, NumPy, TF-IDF, Feature Engineering, Classification, Regression, Model Evaluation

AI: LangChain, RAG, Hugging Face, Mistral AI, Google Gemini, Embeddings, ChromaDB, Whisper, NLP

Web & Backend: React.js, Node.js, Express.js, REST APIs, JWT, MVC Architecture

Databases: MongoDB, Mongoose

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design Basics

Developer Tools: Git, GitHub, Postman, VS Code, Vercel

PROJECTS

Backend Ledger & Transaction Service | *Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, Nodemailer*

- Developed a RESTful backend supporting **authentication, account management, balance tracking and account-to-account transfers**, with APIs designed around modular controllers, services and database models.
- Implemented a **double-entry DEBIT/CREDIT ledger** to maintain an auditable transaction history, separating immutable ledger records from current account balances for traceable financial operations.
- Used **MongoDB transactions and atomic balance updates** to maintain consistency between account balances and ledger entries, with validation for insufficient funds and failed transfers.
- Added **idempotency keys, unique constraints, JWT authentication, bcrypt hashing and centralized error handling** to prevent duplicate transaction processing and improve backend reliability.

AI Interview Preparation Platform | *React.js, Node.js, Express.js, MongoDB, Google Gemini, JWT, Zod, Puppeteer*

- Built a full-stack **AI interview and resume platform** using React.js, Node.js, Express.js and MongoDB for **resume/JD analysis, AI resume generation, personalized interview generation, skill-gap analysis and preparation planning**.
- Integrated **Google Gemini** to analyze candidate profiles against job requirements and generate **ATS-oriented resumes, technical/behavioral questions and interview evaluations**, achieving **94% candidate-job matching accuracy and 90% interview evaluation accuracy**.
- Implemented **JWT authentication with token blacklisting, bcrypt, Zod validation and MongoDB-based interview history**, enabling secure access to previous interview sessions and generated reports.
- Integrated **Puppeteer for dynamic PDF generation** of AI-generated resumes and interview reports, providing downloadable documents directly from the platform

AI Video & Meeting Assistant | *Python, Whisper, Sarvam AI, LangChain, Mistral, ChromaDB, Sentence Transformers, Streamlit, FFmpeg, Pydub, yt-dlp*

- Built an AI-powered meeting assistant that accepts **YouTube URLs and audio/video files**, using **yt-dlp, FFmpeg/Pydub and local Whisper** for English transcription and **Sarvam AI** for Hindi/Hinglish speech-to-text, achieving **10% Word Error Rate (WER)**.
- Developed an automated meeting intelligence pipeline using **Mistral and LangChain** to generate **bullet-point summaries, action items with owners/deadlines, key decisions and unresolved questions** from meeting transcripts.

- Implemented a **RAG-based meeting Q&A system** using Recursive Character Text Splitter, **Sentence Transformer embeddings and ChromaDB** for semantic retrieval, achieving **90% retrieval accuracy and 92% RAG answer accuracy**.
- Built an interactive **Streamlit interface** for transcript analysis and conversational meeting Q&A, with **PDF/TXT report export** for saving generated meeting summaries and insights.

Student Mental Health Score Prediction | *Python, Scikit-learn, Pandas, NumPy, Random Forest, FastAPI, Joblib*

- Built an end-to-end **ML regression system** using lifestyle, academic, social-media and stress features, achieving **0.88 R^2, 0.35 MAE and 0.46 RMSE** with Random Forest.
- Developed a reusable **Scikit-learn preprocessing pipeline** using ColumnTransformer, SimpleImputer, StandardScaler, log transformation, Ordinal/One-Hot Encoding and high-cardinality feature grouping.
- Compared **Linear Regression vs Random Forest** and optimized hyperparameters using **RandomizedSearchCV with 5-fold cross-validation**, then deployed the Joblib pipeline through **FastAPI with Pydantic validation**.

POSITION OF RESPONSIBILITY

Sports Web Coordinator | IIT Hyderabad - Managed the sports web platform, updating event schedules, participant information and sports-related content while coordinating with teams to ensure timely and accurate website updates.

EXPERIENCE

Cybersecurity Intern | Synax

- Conducted **third-party and software supply-chain security assessments** across vendors, SaaS platforms, cloud services, APIs and open-source dependencies, identifying security gaps, vulnerabilities and attack paths.
- Performed **asset discovery, attack-surface analysis, threat modeling, vulnerability assessment and risk prioritization** using **OWASP Top 10 and OWASP API Security Top 10**.
- Used **Nmap, Burp Suite, Wireshark, VirusTotal, vulnerability scanners and MITRE ATT&CK** for network enumeration, web/API security testing, traffic analysis, threat intelligence and attack-technique mapping.
- Investigated **IOCs** including IPs, domains, URLs and file hashes, and assessed software supply-chain/dependency risks using **SCA, CVE/CVSS, EPSS, CISA KEV, SBOM, SLSA and Sigstore**, documenting findings and mitigation recommendations.